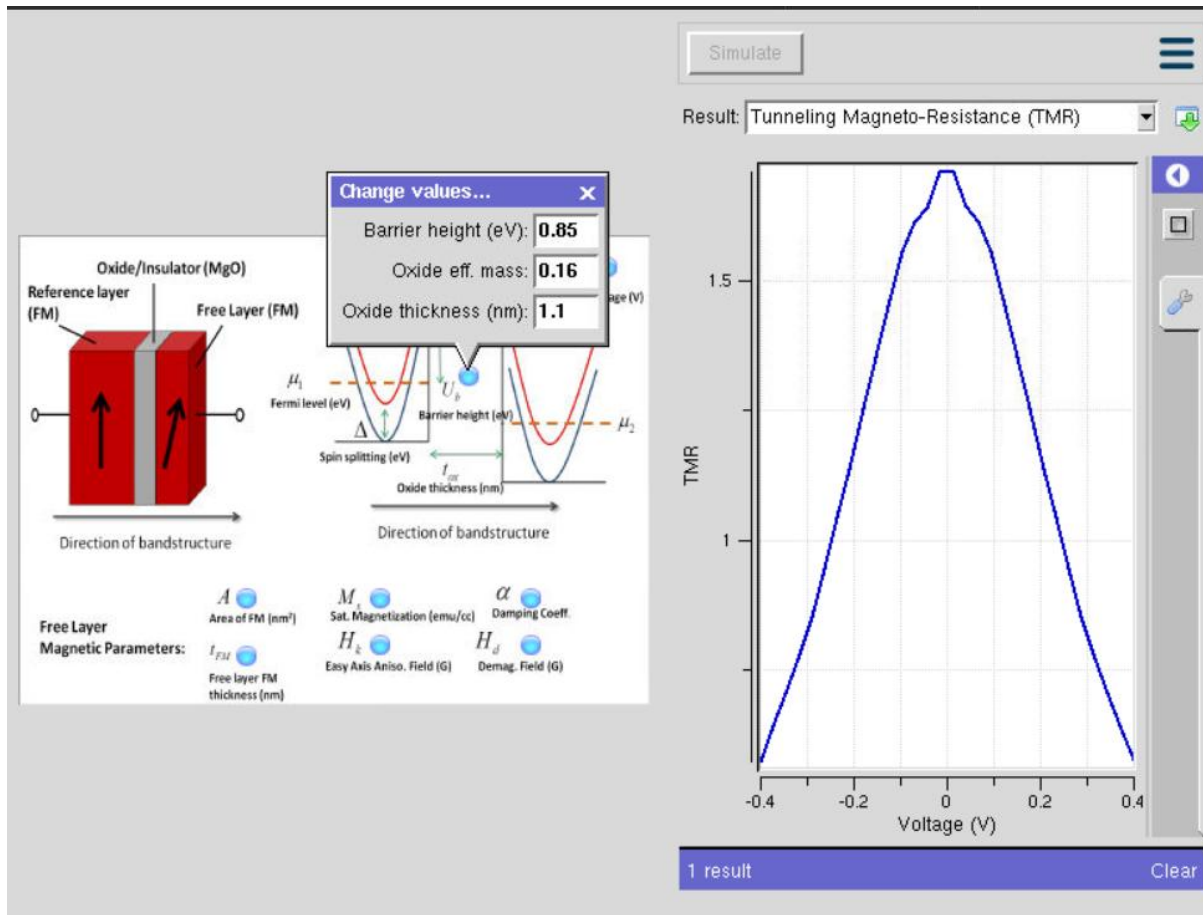


Exp. 9. Simulation of Tunneling Magnetoresistance (TMR) by Varying Barrier Height using NanoHUB

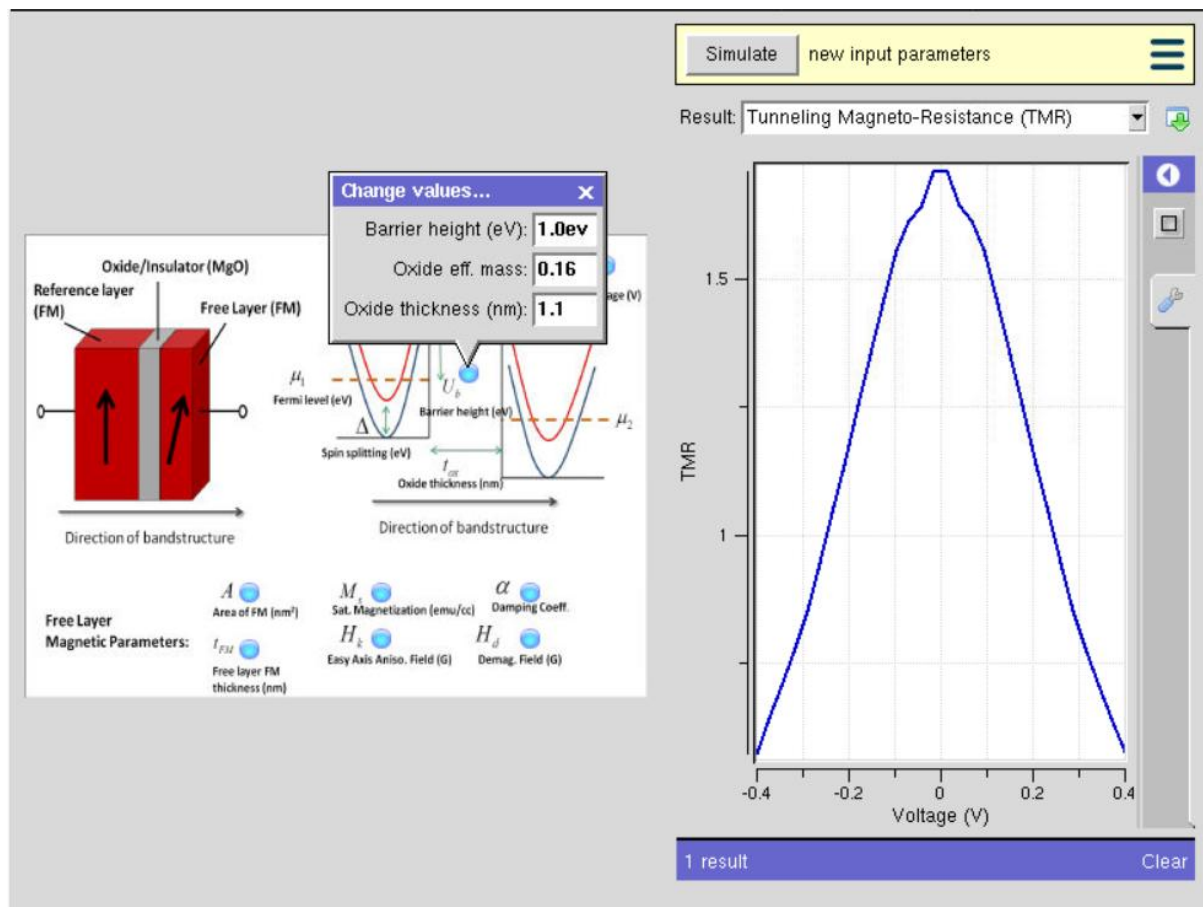
Output:

Case Study 1: Characteristics of TMR with barrier height of 0.85eV



- This figure shows a Magnetic Tunnel Junction (MTJ) with two magnetic layers separated by an MgO insulating layer.
- Electrons tunnel through the thin MgO barrier.
- The tunneling depends on the direction of magnetization of the two magnetic layers.
- This structure is used in spintronic devices such as MRAM and magnetic sensors.

Case Study 2: Characteristics of TMR with barrier height of 1.0eV



- This graph shows TMR versus Voltage.
- TMR is maximum at 0 V.
- As voltage increases in the positive or negative direction, TMR decreases.
- The result indicates that the MTJ works best near zero voltage and shows good spin-dependent tunneling behavior.